National Transportation Safety Board
Aviation Accident Final Report

Location:	Mesa, AZ	**Accident Number:**	LAX08CA031
Date & Time:	11/24/2007, 1600 MST	**Registration:**	N4191D
Aircraft:	Diamond Aircraft Industries DA-20-C1	**Aircraft Damage:**	Substantial
Defining Event:		**Injuries:**	1 None
Flight Conducted Under:	Part 91: General Aviation - Instructional		

Analysis

The student pilot was practicing short-field landings. While on short final in a crab to compensate for the crosswind conditions, he realized that the airplane was below the glide path of the destination runway and began to flare in an effort to reduce the rapid rate of descent. The airplane landed hard on the left main landing gear and began an abrupt veer to the left of runway centerline. The left wing sustained substantial damage. The student pilot reported no preimpact mechanical malfunctions or failures with the airframe or engine.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be: the student pilot's failure to maintain the proper glide path and misjudged flare, which resulted in a hard landing.

Findings

Occurrence #1: HARD LANDING
Phase of Operation: LANDING - FLARE/TOUCHDOWN

Findings
1. WEATHER CONDITION - CROSSWIND
2. (C) PROPER GLIDEPATH - NOT MAINTAINED - PILOT IN COMMAND
3. (C) FLARE - MISJUDGED - PILOT IN COMMAND

Factual Information

Pilot Information

Certificate:	Student	**Age:**	30, Male
Airplane Rating(s):	None	**Seat Occupied:**	
Other Aircraft Rating(s):	None	**Restraint Used:**	Seatbelt, Shoulder harness
Instrument Rating(s):	None	**Second Pilot Present:**	No
Instructor Rating(s):		**Toxicology Performed:**	
Medical Certification:	Class 3	**Last FAA Medical Exam:**	08/01/2007
Occupational Pilot:		**Last Flight Review or Equivalent:**	
Flight Time:	59 hours (Total, all aircraft), 59 hours (Total, this make and model), 59 hours (Last 90 days, all aircraft), 25 hours (Last 30 days, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	Diamond Aircraft Industries	**Registration:**	N4191D
Model/Series:	DA-20-C1	**Aircraft Category:**	Airplane
Year of Manufacture:		**Amateur Built:**	No
Airworthiness Certificate:	Normal	**Serial Number:**	C0408
Landing Gear Type:	Tricycle	**Seats:**	
Date/Type of Last Inspection:		**Certified Max Gross Wt.:**	
Time Since Last Inspection:		**Engines:**	1 Reciprocating
Airframe Total Time:		**Engine Manufacturer:**	Teledyne Continental
ELT:		**Engine Model/Series:**	IO-240-B
Registered Owner:	Sabena Airline Training Center	**Rated Power:**	
Operator:	Sabena Airline Training Center	**Operating Certificate(s) Held:**	None

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual Conditions	**Condition of Light:**	Day
Observation Facility, Elevation:		**Distance from Accident Site:**	
Observation Time:		**Direction from Accident Site:**	
Lowest Cloud Condition:	Clear	**Visibility**	30 Miles
Lowest Ceiling:	Broken / 25000 ft agl	**Visibility (RVR):**	
Wind Speed/Gusts:	8 knots /	**Turbulence Type Forecast/Actual:**	/
Wind Direction:	310°	**Turbulence Severity Forecast/Actual:**	/
Altimeter Setting:	29.96 inches Hg	**Temperature/Dew Point:**	21°C / 2°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Mesa, AZ (FFZ)	**Type of Flight Plan Filed:**	None
Destination:	Mesa, AZ (FFZ)	**Type of Clearance:**	VFR
Departure Time:		**Type of Airspace:**	

Airport Information

Airport:	Falcon Field Airport (FFZ)	**Runway Surface Type:**	Asphalt
Airport Elevation:		**Runway Surface Condition:**	Dry
Runway Used:	4L	**IFR Approach:**	None
Runway Length/Width:	3799 ft / 75 ft	**VFR Approach/Landing:**	Full Stop; Traffic Pattern

Wreckage and Impact Information

Crew Injuries:	1 None	**Aircraft Damage:**	Substantial
Passenger Injuries:	N/A	**Aircraft Fire:**	None
Ground Injuries:	N/A	**Aircraft Explosion:**	
Total Injuries:	1 None	**Latitude, Longitude:**	33.460833, -111.728333

Administrative Information

Investigator In Charge (IIC):	Zoë Keliher	**Report Date:**	01/31/2008
Additional Participating Persons:	Stacy Skrocki; Federal Aviation Administration; Scottsdale, AZ		
Publish Date:			
Note:	This accident report documents the factual circumstances of this accident as described to the NTSB.		
Investigation Docket:	NTSB accident and incident dockets serve as permanent archival information for the NTSB's investigations. Dockets released prior to June 1, 2009 are publicly available from the NTSB's Record Management Division at pubinq@ntsb.gov, or at 800-877-6799. Dockets released after this date are available at http://dms.ntsb.gov/pubdms/.		

The National Transportation Safety Board (NTSB), established in 1967, is an independent federal agency mandated by Congress through the Independent Safety Board Act of 1974 to investigate transportation accidents, determine the probable causes of the accidents, issue safety recommendations, study transportation safety issues, and evaluate the safety effectiveness of government agencies involved in transportation. The NTSB makes public its actions and decisions through accident reports, safety studies, special investigation reports, safety recommendations, and statistical reviews.

The Independent Safety Board Act, as codified at 49 U.S.C. Section 1154(b), precludes the admission into evidence or use of any part of an NTSB report related to an incident or accident in a civil action for damages resulting from a matter mentioned in the report. A factual report that may be admissible under 49 U.S.C. § 1154(b) is available here.